

ECE 0102 Microelectronic Circuits

Chapter 3 Problem Solutions: Solid-State Diodes and Diode Circuits

Generated study solution set

Constants used unless otherwise stated: $q = 1.602 \times 10^{-19}$ C, $\epsilon_s = 11.8\epsilon_0 = 1.045 \times 10^{-12}$ F/cm, $n_i = 10^{10}$ cm⁻³ at 300 K, $V_T = kT/q \approx 0.025$ V at room temperature. For silicon abrupt junctions,

$$\phi_j = V_T \ln \frac{N_A N_D}{n_i^2}, \quad W_d = \sqrt{\frac{2\epsilon_s}{q} \frac{N_A + N_D}{N_A N_D} (\phi_j + V_R)},$$

$$x_p = \frac{N_D}{N_A + N_D} W_d, \quad x_n = \frac{N_A}{N_A + N_D} W_d, \quad E_{max} = \frac{2(\phi_j + V_R)}{W_d}.$$

Scope note. This PDF gives hand-solution results and procedures for the whole Chapter 3 problem set shown. Problems explicitly asking for MATLAB, spreadsheet, SPICE, or graph plotting are solved as numerical/setup procedures; simulation plots are not fabricated. Several later multi-diode graph-reading problems require the exact printed characteristic or simulator, so the operating equations and decision logic are provided.

3.1 pn Junction Diode

Problem 3.1

Given $N_A = 10^{19}$, $N_D = 10^{18}$ cm⁻³.

$$\phi_j = 0.025 \ln(10^{37}/10^{20}) = 0.979 \text{ V.}$$

$$W_{d0} = 3.75 \times 10^{-6} \text{ cm} = 0.0375 \text{ }\mu\text{m.}$$

$$x_p = \frac{10^{18}}{1.1 \times 10^{19}} W = 0.00341 \text{ }\mu\text{m}, \quad x_n = 0.0341 \text{ }\mu\text{m.}$$

$$E_{max} = \frac{2\phi_j}{W} = 5.22 \times 10^5 \text{ V/cm.}$$

Problem 3.2

$N_A = 10^{18}$, $N_D = 10^{15}$. On the p side: $p_p \approx N_A = 10^{18}$ and $n_p = n_i^2/N_A = 10^2$ cm⁻³. On the n side: $n_n \approx N_D = 10^{15}$ and $p_n = n_i^2/N_D = 10^5$ cm⁻³.

$$\phi_j = 0.748 \text{ V}, \quad W_{d0} = 9.88 \times 10^{-5} \text{ cm} = 0.988 \text{ }\mu\text{m.}$$

Since the n side is lightly doped, almost the entire depletion region lies in the n side: $x_p = 0.000987 \text{ }\mu\text{m}$, $x_n = 0.987 \text{ }\mu\text{m}$.

Problem 3.3

Repeat 3.2 with $N_A = 10^{16}$, $N_D = 10^{19}$.

$$p_p \approx 10^{16}, \quad n_p = 10^4, \quad n_n \approx 10^{19}, \quad p_n = 10.$$

$$\phi_j = 0.863 \text{ V}, \quad W_{d0} = 3.36 \times 10^{-5} \text{ cm} = 0.336 \text{ }\mu\text{m.}$$

Most of the depletion layer lies on the p side: $x_p = 0.335 \text{ }\mu\text{m}$, $x_n = 3.35 \times 10^{-4} \text{ }\mu\text{m}$.

Problem 3.4

$N_A = N_D = 10^{18}$.

$$p_p = n_n = 10^{18}, \quad n_p = p_n = 10^2.$$

$$\phi_j = 0.921 \text{ V}, \quad W_{d0} = 4.90 \times 10^{-6} \text{ cm} = 0.0490 \text{ }\mu\text{m.}$$

Symmetric junction: $x_p = x_n = 0.0245 \text{ }\mu\text{m}$.

Problem 3.5

$N_D = 10^{20}$, $N_A = 10^{18}$.

$$p_p \approx 10^{18}, \quad n_p = 10^2, \quad n_n \approx 10^{20}, \quad p_n = 1.$$

$$\phi_j = 1.04 \text{ V}, \quad W_{d0} = 3.69 \times 10^{-6} \text{ cm} = 0.0369 \text{ }\mu\text{m.}$$

$$x_p = 0.0366 \text{ }\mu\text{m}, \quad x_n = 3.66 \times 10^{-4} \text{ }\mu\text{m.}$$

Problem 3.6

For an abrupt junction,

$$W_d = W_{d0} \sqrt{\frac{\phi_j + V_R}{\phi_j}}.$$

(a) To triple width, $3 = \sqrt{(\phi_j + V_R)/\phi_j}$, so

$$V_R = (9 - 1)\phi_j = 8(0.85) = 6.8 \text{ V}.$$

(b) For $V_R = 7 \text{ V}$,

$$W_d = 0.4 \sqrt{\frac{0.85 + 7}{0.85}} = 1.22 \mu\text{m}.$$

Problem 3.7

(a) To double width,

$$V_R = (4 - 1)\phi_j = 3(0.6) = 1.8 \text{ V}.$$

(b) With $V_R = 12 \text{ V}$,

$$W_d = 1 \sqrt{\frac{12.6}{0.6}} = 4.58 \mu\text{m}.$$

Problem 3.8

In a neutral region with resistivity $\rho = 0.5 \Omega \text{ cm}$,

$$J = \sigma E = \frac{E}{\rho} \Rightarrow E = J\rho = 2000(0.5) = 1000 \text{ V/cm}.$$

Problem 3.9

Similarly,

$$E = J\rho = 5000(2.5) = 1.25 \times 10^4 \text{ V/cm}.$$

Problem 3.10

For p-type silicon, majority holes dominate:

$$J_{max} = qp v_{max} \approx (1.602 \times 10^{-19})(4 \times 10^{17})(10^7) = 6.41 \times 10^5 \text{ A/cm}^2.$$

Problem 3.11

For n-type silicon,

$$J_{max} = qn v_{max} = (1.602 \times 10^{-19})(5 \times 10^{15})(10^7) = 8.01 \times 10^3 \text{ A/cm}^2.$$

Problem 3.12

Given $N_A(x) = N_0 e^{-x/L}$ and $p(x) = N_A(x)$ in thermal equilibrium with $J_p = 0$:

$$J_p = qp\mu_p E - qD_p \frac{dp}{dx} = 0.$$

Since $dp/dx = -p/L$,

$$qp\mu_p E + qD_p \frac{p}{L} = 0 \Rightarrow E = -\frac{D_p}{\mu_p L} = -\frac{V_T}{L}.$$

For $L = 1 \mu\text{m} = 10^{-4} \text{ cm}$,

$$E = -\frac{0.025}{10^{-4}} = -250 \text{ V/cm}.$$

It is independent of N_0 .

Problem 3.13

For electron diffusion,

$$J_{n,diff} = qD_n \frac{dn}{dx}, \quad D_n = \mu_n V_T = 500(0.025) = 12.5 \text{ cm}^2/\text{s}.$$

Required gradient:

$$\frac{dn}{dx} = \frac{J_n}{qD_n} = \frac{2000}{(1.602 \times 10^{-19})(12.5)} = 9.99 \times 10^{20} \text{ cm}^{-4}.$$

Problem 3.14

Solve Eq. (3.25) with $I_S = 10^{-16}$ A using calculator solver. Use the diode/load-line equation from the text. Input the equation as $f(V_D) = 0$ and solve near 0.7 V. Report the positive root. This is a numerical calculator exercise; the exact answer depends on the circuit constants in Eq. (3.25).

Problem 3.15

Same as 3.14 but with $I_S = 10^{-13}$ A. Use spreadsheet iteration: start with $V_D = 0.7$ V, compute diode current, update from the load equation, and repeat until $|\Delta V_D| < 10^{-4}$ V.

Problem 3.16

Same numerical solution with MATLAB/MATHCAD. Use a root finder such as **fzero**. The root is the operating diode voltage; current follows from the load line.

3.2-3.4 Diode Equation and Bias

Problem 3.17

Thermal voltage:

$$V_T = \frac{kT}{q}.$$

If $V_T = 0.025$ V, then

$$T = \frac{qV_T}{k} = 290 \text{ K} = 16.9^\circ\text{C}.$$

At -55°C , $T = 218 \text{ K}$, $V_T = 18.8 \text{ mV}$. At 0°C , $T = 273 \text{ K}$, $V_T = 23.5 \text{ mV}$. At $+85^\circ\text{C}$, $T = 358 \text{ K}$, $V_T = 30.9 \text{ mV}$.

Problem 3.18

Diode equation:

$$i_D = I_S \left(e^{v_D/(nV_T)} - 1 \right).$$

(a) Plot for $I_S = 10^{-12}$, $n = 1$. (b) Repeat with $n = 2$; the curve rises more slowly versus v_D . (c) Repeat with $I_S = 10^{-14}$; for the same current, the voltage increases by approximately $nV_T \ln(100)$.

Problem 3.19

Given $n = 1.05$, $T = 320 \text{ K}$:

$$nV_T = n \frac{kT}{q} = 1.05(8.617 \times 10^{-5})(320) = 0.02895 \text{ V}.$$

For $n = 1.00$ with same nV_T ,

$$T = \frac{0.02895}{8.617 \times 10^{-5}} = 336 \text{ K}.$$

Problem 3.20

Use the reverse-bias model Eq. (3.19). With $I_{S0} = 11 \text{ fA}$ and $\phi_j = 0.8 \text{ V}$, plot i_D over $-10 \leq v_D \leq 0$. The curve remains near $-I_S$ until close to the breakdown model limit specified by Eq. (3.19).

Problem 3.21

From the semilog straight line,

$$\log_{10} I_D = \log_{10} I_S + \frac{v_D}{nV_T \ln 10}.$$

Use two points on Fig. P3.21. Slope $m = \Delta \log_{10} I / \Delta V$ gives

$$n = \frac{1}{mV_T \ln 10}, \quad I_S = I_D e^{-v_D/(nV_T)}.$$

Reading approximate points from the printed plot gives $n \approx 1$ and I_S on the order of 10^{-13} to 10^{-14} A depending on the graph reading.

Problem 3.22

$I_S = 10^{-17}$, $n = 1.07$.

$$V_D = nV_T \ln(1 + I_D/I_S).$$

(a) $I_D = 70 \mu\text{A}$: $V_D = 0.791 \text{ V}$. (b) $I_D = 5 \mu\text{A}$: $V_D = 0.721 \text{ V}$. (c) $v_D = 0$: $i_D = 0$. (d) $v_D = -0.075 \text{ V}$: $i_D = -9.39 \times 10^{-18} \text{ A}$. (e) $v_D = -5 \text{ V}$: $i_D \approx -I_S = -10^{-17} \text{ A}$.

Problem 3.23

$I_S = 10^{-18}$, $n = 1$. (a) $I_D = 100\mu A$: $V_D = 0.806$ V. (b) $I_D = 10\mu A$: $V_D = 0.748$ V. (c) $i_D(0) = 0$. (d) $v_D = -0.06$ V: $i_D = -9.09 \times 10^{-19}$ A. (e) $v_D = -4$ V: $i_D \approx -10^{-18}$ A.

Problem 3.24

$I_S = 10^{-16}$, $n = 1$. (a)

$$i_D = 10^{-16}(e^{0.675/0.025} - 1) = 5.32 \times 10^{-5} A = 53.2\mu A.$$

(b) If current triples, voltage increases by

$$\Delta V_D = nV_T \ln 3 = 27.5 mV.$$

So $V_D \approx 0.702$ V.

Problem 3.25

$I_S = 10^{-10}$, $n = 2$. (a) $I_D = 40$ A:

$$V_D = 2(0.025) \ln(1 + 40/10^{-10}) = 1.336 \text{ V}.$$

(b) $I_D = 100$ A: $V_D = 1.382$ V.

Problem 3.26

$i_D = 300\mu A$, $v_D = 0.75$ V, $n = 1$.

$$I_S = \frac{I_D}{e^{v_D/V_T} - 1} = 2.81 \times 10^{-17} A.$$

For $v_D = -3$ V, $i_D \approx -I_S = -2.81 \times 10^{-17}$ A.

Problem 3.27

$i_D = 2$ mA, $v_D = 0.82$ V, $n = 1$.

$$I_S = \frac{2 \times 10^{-3}}{e^{0.82/0.025} - 1} = 1.14 \times 10^{-17} A.$$

For $v_D = -5$ V, $i_D \approx -1.14 \times 10^{-17}$ A.

Problem 3.28

For $I_D = 1$ mA,

$$V_D = nV_T \ln(1 + I_D/I_S).$$

With $I_S = 10^{-14}$, $V_D = 0.633$ V; with $I_S = 10^{-12}$, $V_D = 0.518$ V. Range: 0.518 to 0.633 V.

Problem 3.29

Use temperature dependence of diode voltage at fixed bias. If current doubles at fixed 0.9 V, find T_2 from

$$\frac{I_2}{I_1} \approx \frac{I_S(T_2)}{I_S(T_1)} \exp \left[\frac{V_D}{n} \left(\frac{1}{V_T(T_2)} - \frac{1}{V_T(T_1)} \right) \right]$$

or use the text's temperature coefficient method if supplied. As a fast approximation, silicon diode current roughly doubles for a $10^\circ C$ rise at fixed voltage. Thus (a) about 325 K, (b) about 305 K.

Problem 3.30

Fit the data to

$$\ln I_D = \ln I_S + \frac{V_D}{nV_T}.$$

At $T = 307$ K, $V_T = 0.02646$ V. Linear regression of $\ln I_D$ versus V_D gives slope $1/(nV_T)$ and intercept $\ln I_S$. Use all table points for least squares. The reported M is the sum of squared current residuals, not log residuals; refine I_S, n by nonlinear least squares after the linear estimate.

3.5 Temperature Coefficient

Problem 3.31

$V_T = kT/q$. For $-40^\circ C$, $T = 233$ K, $V_T = 20.1$ mV. For $0^\circ C$, $T = 273$ K, $V_T = 23.5$ mV. For $50^\circ C$, $T = 323$ K, $V_T = 27.8$ mV.

Problem 3.32

$I_S = 10^{-15}$, $n = 1$, $I_D = 100\mu A$ at $25^\circ C$.

$$V_D = 0.025 \ln(10^{11}) = 0.633 \text{ V.}$$

At $50^\circ C$, $\Delta T = 25K$ and coefficient $-1.8mV/K$:

$$V_D \approx 0.633 - 0.0018(25) = 0.588 \text{ V.}$$

Problem 3.33

$I_S = 2.5 \times 10^{-16}$ at $30^\circ C$, $I_D = 1mA$. Using $V_T \approx k(303)/q = 0.0261V$,

$$V_D = 0.0261 \ln(1 + 10^{-3}/2.5 \times 10^{-16}) = 0.760 \text{ V.}$$

At $50^\circ C$ with coefficient $-2mV/K$,

$$V_D = 0.760 - 0.002(20) = 0.720 \text{ V.}$$

Problem 3.34

$I_S = 10^{-15}$, $n = 1$, $I_D = 250\mu A$. At $25^\circ C$,

$$V_D = 0.025 \ln(2.5 \times 10^{11}) = 0.656 \text{ V.}$$

At $85^\circ C$, relative to $55^\circ C$ coefficient $-2mV/K$ is used as an approximation. From $25^\circ C$ to $85^\circ C$, $\Delta T = 60K$:

$$V_D \approx 0.656 - 0.120 = 0.536 \text{ V.}$$

Problem 3.35

With $I_S = CT^3 e^{-E_G/kT}$ and $I_D \approx I_S e^{V_D/V_T}$ at fixed current,

$$0 = d \ln I_D = d \ln I_S + d \left(\frac{V_D}{V_T} \right).$$

This gives

$$\frac{dV_D}{dT} \approx \frac{V_D}{T} - \frac{3V_T}{T} - \frac{E_G}{qT}.$$

At $E_G = 1.21eV$, $V_D = 0.7V$, $T = 300K$, result is approximately $-2.2mV/K$.

Problem 3.36

Since $I_S \propto T^3 e^{-E_G/kT}$, solve

$$\ln \frac{I_S(T_2)}{I_S(T_1)} = 3 \ln \frac{T_2}{T_1} - \frac{E_G}{k} \left(\frac{1}{T_2} - \frac{1}{T_1} \right).$$

Use $T_1 = 300K$. Set the left side to $\ln 2$, $\ln 10$, or $-\ln 100$ and solve numerically for T_2 .

3.6 Reverse Bias and Breakdown

Problem 3.37

Use

$$W_d = W_{d0} \sqrt{\frac{\phi_j + V_R}{\phi_j}}.$$

(a) With $W_{d0} = 1\mu m$, $\phi_j = 0.8V$, $V_R = 5V$,

$$W_d = \sqrt{5.8/0.8} = 2.69 \mu m.$$

(b) $V_D = -10V$ means $V_R = 10V$:

$$W_d = \sqrt{10.8/0.8} = 3.67 \mu m.$$

Problem 3.38

$N_D = 10^{20}$, $N_A = 10^{18}$. From 3.5, $W_{d0} = 0.0369\mu m$ and $\phi_j = 1.04V$. At $V_R = 5V$,

$$W_d = 0.0369 \sqrt{6.04/1.04} = 0.0889 \mu m.$$

At $V_R = 25V$,

$$W_d = 0.0369 \sqrt{26.04/1.04} = 0.184 \mu m.$$

Problem 3.39

$$N_D = 10^{15}, N_A = 10^{16}.$$

$$\phi_j = 0.025 \ln(10^{31}/10^{20}) = 0.633V, \quad W_{d0} \approx 0.899 \mu\text{m}.$$

For $V_R = 10V$,

$$W_d = 0.899 \sqrt{10.633/0.633} = 3.69 \mu\text{m}.$$

For $V_R = 100V$,

$$W_d = 0.899 \sqrt{100.633/0.633} = 11.3 \mu\text{m}.$$

Problem 3.40

Breakdown when $E_{max} = 300kV/cm$. Since $E = 2(\phi + V_R)/W$ and $W = W_0 \sqrt{(\phi + V_R)/\phi}$,

$$E = \frac{2\sqrt{\phi(\phi + V_R)}}{W_0}.$$

Solve:

$$\phi + V_R = \frac{(EW_0/2)^2}{\phi}.$$

With $W_0 = 1 \mu\text{m} = 10^{-4}cm$, $\phi = 0.6V$:

$$\phi + V_R = \frac{(300000 \cdot 10^{-4}/2)^2}{0.6} = 375V, \quad V_R \approx 374V.$$

Problem 3.41

Use the junction of Prob. 3.2: $W_0 = 0.988 \mu\text{m}$, $\phi = 0.748V$. With $E_{crit} = 300kV/cm$,

$$\phi + V_R = \frac{(EW_0/2)^2}{\phi} \approx 439V, \quad V_R \approx 438V.$$

Problem 3.42

From Fig. P3.42, the Zener characteristic is vertical at approximately $v_D = -4V$. Thus $V_Z \approx 4V$. Since the line is drawn vertical, the small-signal Zener resistance is approximately

$$R_Z \approx 0\Omega.$$

Problem 3.43

For one-sided abrupt junction $N_A \gg N_D$, breakdown is on the lightly doped side:

$$E_{crit}^2 \approx \frac{2qN_D(V_{BR} + \phi_j)}{\epsilon_s}.$$

For $V_{BR} \approx 1000V$, $\phi_j \ll V_{BR}$,

$$N_D \approx \frac{\epsilon_s E_{crit}^2}{2qV_{BR}} = \frac{(1.045 \times 10^{-12})(3 \times 10^5)^2}{2(1.602 \times 10^{-19})(1000)} = 2.94 \times 10^{14} \text{ cm}^{-3}.$$

3.7 Junction Capacitance

Problem 3.44

Junction capacitance per area:

$$C'_j = \frac{\epsilon_s}{W_d}.$$

For $N_A = 10^{18}$, $N_D = 10^{15}$, $W_0 = 0.988 \mu\text{m}$:

$$C'_{j0} = \frac{1.045 \times 10^{-12}}{9.88 \times 10^{-5}} = 1.06 \times 10^{-8} \text{ F/cm}^2.$$

At $V_R = 9V$,

$$C_j = C_{j0} \sqrt{\frac{\phi}{\phi + V_R}}.$$

With $A = 0.02 \text{ cm}^2$, $C \approx 0.02(1.06 \times 10^{-8}) \sqrt{0.748/9.748} = 5.87 \times 10^{-11} \text{ F} = 58.7 \text{ pF}$.

Problem 3.45

$N_A = 10^{15}$, $N_D = 10^{20}$ is one-sided. Compute W_0 and ϕ_j as in 3.1:

$$\phi_j \approx 0.863V, \quad W_0 \approx 1.06 \mu\text{m}.$$

$$C'_{j0} \approx 9.9 \times 10^{-9} F/\text{cm}^2.$$

At $V_R = 3V$ and $A = 0.05\text{cm}^2$:

$$C \approx AC'_{j0} \sqrt{\phi/(\phi+3)} \approx 0.05(9.9 \times 10^{-9}) \sqrt{0.863/3.863} = 2.34 \times 10^{-10} F = 234\text{pF}.$$

Problem 3.46

Diffusion capacitance:

$$C_d = \tau_T \frac{dI}{dV} = \tau_T \frac{I_D}{nV_T} \approx \frac{\tau_T I_D}{V_T}.$$

(a) $I_D = 200\mu A$, $\tau_T = 100\text{ps}$:

$$C_d = \frac{10^{-10}(2 \times 10^{-4})}{0.025} = 8.0 \times 10^{-13} F = 0.80\text{pF}.$$

(b) Stored charge $Q = \tau_T I_D = 2.0 \times 10^{-14} C$. (c) For $I_D = 5\text{mA}$, $C_d = 20\text{pF}$, $Q = 5.0 \times 10^{-13} C$.

Problem 3.47

(a) $I_D = 1A$, $\tau_T = 10\text{ns}$:

$$C_d = \frac{10^{-8}(1)}{0.025} = 4.0 \times 10^{-7} F = 0.40\mu F.$$

(b) $Q = 10^{-8} C$. (c) For $I_D = 100\text{mA}$, $C_d = 40\text{nF}$, $Q = 10^{-9} C$.

Problem 3.48

Square diode side $5\text{mm} = 0.5\text{cm}$, area $A = 0.25\text{cm}^2$. $N_A = 10^{19}$, $N_D = 10^{16}$. One-sided width is close to that set by N_D . Use $\phi_j = 0.025 \ln(10^{35}/10^{20}) = 0.863V$.

$$W_0 \approx 0.335 \mu\text{m}, \quad C_0 = A\epsilon_s/W_0 \approx 7.8\text{nF}.$$

At $V_R = 4V$,

$$C = C_0 \sqrt{\phi/(\phi+4)} \approx 3.29\text{nF}.$$

Problem 3.49

Area $A = 10^4 \mu\text{m}^2 = 10^{-4} \text{cm}^2$. $N_A = 10^{19}$, $N_D = 10^{17}$.

$$\phi_j = 0.025 \ln(10^{36}/10^{20}) = 0.921V, \quad W_0 \approx 0.108 \mu\text{m}.$$

$$C_0 = A\epsilon_s/W_0 \approx 9.7\text{pF}.$$

At $V_R = 5V$,

$$C = C_0 \sqrt{0.921/5.921} \approx 3.8\text{pF}.$$

Problem 3.50

Varactor:

$$C_j = C_{j0} \sqrt{\frac{\phi_j}{\phi_j + V_{DC}}}, \quad f_0 = \frac{1}{2\pi\sqrt{LC}}.$$

With $C_{j0} = 39\text{pF}$, $\phi_j = 0.8V$, $L = 10\mu H$: For $V_{DC} = 1V$, $C = 39\sqrt{0.8/1.8} = 26.0\text{pF}$, $f_0 \approx 9.87\text{MHz}$. For $V_{DC} = 9V$, $C = 39\sqrt{0.8/9.8} = 11.1\text{pF}$, $f_0 \approx 15.1\text{MHz}$.

3.8 Schottky Barrier Diode

Problem 3.51

Schottky equation with $I_S = 10^{-11} A$, $n = 1$. (a) For $I = 4\text{mA}$,

$$V_D = 0.025 \ln(1 + 4 \times 10^{-3}/10^{-11}) = 0.495V.$$

(b) pn diode with $I_S = 10^{-14} A$ at same current:

$$V_D = 0.025 \ln(4 \times 10^{11}) = 0.668V.$$

Problem 3.52

(a) Schottky $I_S = 10^{-7} A$, $I = 50 A$:

$$V_D = 0.025 \ln(5 \times 10^8) = 0.501 V.$$

(b) pn diode $I_S = 10^{-15} A$, $n = 2$:

$$V_D = 0.05 \ln(5 \times 10^{16}) = 1.92 V.$$

3.9 SPICE Model and Layout

Problem 3.53

Series resistance adds ohmic drop:

$$V'_D = V_D + I_D R_S.$$

For $I_S = 5 \times 10^{-16} A$, $I_D = 1 mA$:

$$V_D = 0.025 \ln(1 + 10^{-3}/5 \times 10^{-16}) = 0.708 V.$$

(a) $R_S = 10 \Omega$: $V'_D = 0.708 + 0.010 = 0.718 V$. (b) $R_S = 100 \Omega$: $V'_D = 0.808 V$.

Problem 3.54

Series resistance of neutral regions:

$$R_S = \rho_p \frac{L_p}{A} + \rho_n \frac{L_n}{A}.$$

$L_p = L_n = 250 \mu m = 0.025 cm$, $A = 0.01 cm^2$.

$$R_p = 2 \frac{0.025}{0.01} = 5 \Omega, \quad R_n = 0.01 \frac{0.025}{0.01} = 0.025 \Omega.$$

$$R_S = 5.03 \Omega.$$

Problem 3.55

Contact resistance:

$$R_c = \frac{\rho_c}{A_c}.$$

Given $\rho_c = 10 \Omega \mu m^2$. For $1 \mu m \times 1 \mu m$, $A = 1 \mu m^2$, so each contact is

$$R_c = 10 \Omega.$$

Total anode+cathode contact resistance is 20Ω if one contact per terminal.

Problem 3.56

(a) If each contact is $1 \mu m^2$, diode active area estimate is set by the two contacts plus separation/guard geometry in Fig. 3.21(a). Minimum estimate: about $2 \mu m^2$ plus spacing. (b) For $0.13 \mu m \times 0.13 \mu m$, each contact area is $0.0169 \mu m^2$, so each contact resistance is

$$R_c = 10/0.0169 \approx 592 \Omega.$$

This shows contact scaling can dominate resistance.

3.10 Load-Line and Circuit Models

Problem 3.57–3.61

For Fig. P3.57, load line is

$$i_D = \frac{V - v_D}{R}.$$

Plot this on the selected diode characteristic. The Q-point is the intersection. For ideal/CVD estimates: if forward biased, ideal: $v_D = 0$, $i_D = V/R$; CVD: $v_D \approx 0.6$ or $0.7 V$, $i_D = (V - V_{on})/R$ if positive; reverse biased gives $i_D \approx 0$ unless Zener breakdown is reached in Fig. P3.42.

Problem 3.62–3.67

Iterative mathematical model: solve

$$V = R i_D + n V_T \ln(1 + i_D/I_S)$$

for i_D , or equivalently solve

$$f(V_D) = V - R I_S (e^{V_D/n V_T} - 1) - V_D = 0.$$

Use direct trial, spreadsheet iteration, or calculator/MATLAB solver. Convergence issues occur when the starting current is 0 or when the exponential overflows; use $V_D \approx 0.6$ – $0.8 V$ as a robust initial guess.

Problem 3.68–3.73

Ideal and CVD diode models: replace an on diode by a short circuit for ideal model, or by a fixed drop V_{on} for CVD model. Replace an off diode by an open circuit. Then verify consistency: on diodes must carry positive current; off diodes must have reverse/non-forward voltage. For Fig. P3.69, first compute node voltages with diode open; if the diode is forward biased under that assumption, turn it on and solve with $v_D = 0$ or $0.6V$.

3.11 Multiple Diode Circuits

Problem 3.74–3.80

Procedure for each multiple-diode circuit: (1) assume each diode state; (2) replace each ON diode by a short or V_{on} source; (3) solve the resulting resistor network; (4) reject any state with negative on-current or forward voltage across an off diode. For Fig. P3.74 and Fig. P3.76, this systematic state testing is safer than guessing by visual inspection. Simulation-only problems 3.78 and 3.80 require SPICE; compare the diode currents and terminal voltages to the ideal/CVD approximations.

3.12 Breakdown Region

Problem 3.81

For Fig. P3.81, replace Zener in breakdown by $V_Z = 4V$, $R_Z = 0$. The output node is clamped near $4V$ if Zener current is positive. Source current:

$$I_S = \frac{24 - 4}{10k} = 2.0mA.$$

Load current:

$$I_L = \frac{4}{3.6k} = 1.11mA.$$

Zener current:

$$I_Z = 0.889mA > 0.$$

Q-point: $v_D = -4V$, $i_D \approx -0.889mA$ using diode current sign convention.

Problem 3.82

Same Q-point as 3.81 by direct circuit analysis: $V_o = 4V$, $I_Z = 0.889mA$.

Problem 3.83

Regulation maintained when $I_Z > 0$:

$$I_S = \frac{30 - 9}{15k} = 1.4mA.$$

Maximum load current is $I_{L,max} \approx 1.4mA$.

$$R_{L,min} = \frac{9V}{1.4mA} = 6.43k\Omega.$$

Problem 3.84

For $R_L = \infty$, all source current flows in Zener:

$$I_Z = 1.4mA, \quad P_Z = V_Z I_Z = 9(1.4mA) = 12.6mW.$$

Problem 3.85

Nominal $R_L = 10k\Omega$: $I_L = 9/10k = 0.9mA$, so

$$I_Z = 1.4 - 0.9 = 0.5mA, \quad P_Z = 4.5mW.$$

Worst case: maximize/minimize using V_S, V_Z, R, R_L tolerances. Use

$$I_Z = \frac{V_S - V_Z}{R} - \frac{V_Z}{R_L}, \quad P_Z = V_Z I_Z.$$

Max I_Z : V_S high, V_Z low, R low, R_L high. Min I_Z : opposite.

Problem 3.86

Fig. P3.86: $V_S = 50V$, $R = 150\Omega$, $V_Z = 15V$. (a) $R_L = 100\Omega$: load current would be $15/100 = 150mA$; source current is $(50 - 15)/150 = 233mA$; $I_Z = 83.3mA$. $P_Z = 1.25W$. (b) $R_L = \infty$: $I_Z = 233mA$, $P_Z = 3.50W$.

Problem 3.87

Nominal with $R_L = 100\Omega$ from 3.86: $I_Z = 83.3mA$, $P_Z = 1.25W$. Worst-case use

$$I_Z = \frac{V_S - V_Z}{R} - \frac{V_Z}{R_L}$$

with $\pm 10\%$ on all quantities. Max: V_S high, V_Z low, R low, R_L high. Min: V_S low, V_Z high, R high, R_L low. Then multiply by worst-case V_Z for P_Z bounds.

3.13 Half-Wave Rectifiers

Problem 3.88

$I_S = 10^{-9}$, $n = 2$, peak current $48.6A$:

$$V_D = 2(0.025) \ln(1 + 48.6/10^{-9}) = 1.23V.$$

Problem 3.89

$I_S = 10^{-8}$, $n = 1.6$, $I = 100A$:

$$V_D = 1.6(0.025) \ln(10^{10}) = 0.921V.$$

Peak diode loss includes junction drop and series resistance. Approx pulse average over one period with conduction time $t_c = 1ms$ at $60Hz$:

$$P \approx (V_D I + I^2 R_s) \frac{t_c}{T} = (0.921 \cdot 100 + 100^2 \cdot 0.01) \frac{0.001}{1/60} \approx 11.5W.$$

Problem 3.90

Numerical conduction angle equation:

$$[(V_P - V_{on})e^{-t/RC} = V_P \cos \omega t - V_{on}]$$

should be solved for the smallest positive conduction start time near the peak. Compare the resulting ripple with the approximate Eq. (3.57). Beware multiple mathematical roots.

Problem 3.91

If ripple is triangular/sawtooth-like, average dc value is approximately

$$V_{dc} = V_{max} - \frac{V_r}{2}.$$

Given $V_r = 0.05(V_P - V_{on})$ and $V_P - V_{on} = 18V$,

$$V_r = 0.9V, \quad V_{dc} = 18 - 0.45 = 17.55V.$$

Problem 3.92

Negative rectifier waveform is the sign-inverted version of the positive output: capacitor charges to approximately $-(V_P - V_{on})$ and discharges toward zero between peaks. Ripple polarity is negative.

Problem 3.93

Substitute the half-wave capacitor-discharge relation into Eq. (3.61) and simplify the exponential discharge over one period; the algebra gives Eq. (3.62). This is a derivation/proof exercise.

Problem 3.94

$v_{I,rms} = 12.6V \pm 10\%$, so $V_P = \sqrt{2}v_{rms}$. Nominal peak is $17.82V$. With $V_{on} = 1V$, ideal capacitor output approximately

$$V_O \approx V_P - V_{on} = 16.82V$$

minus half the ripple if ripple is included. Worst case uses $v_{rms} = 11.34V$ and $13.86V$: output peak range $15.04V$ to $18.60V$ before ripple averaging.

Problem 3.95

$v_{rms} = 6.3V$, so $V_P = 8.91V$. With $V_{on} = 1V$, (a) $V_O \approx 7.91V$ minus ripple average. (b) For half-wave,

$$V_r \approx \frac{I_L}{fC}, \quad I_L \approx \frac{V_O}{R}.$$

Use $R = 0.5\Omega$, $V_r < 0.25V$, $f = 60Hz$:

$$C > \frac{I_L}{fV_r} \approx \frac{7.91/0.5}{60(0.25)} = 1.05F.$$

(c) PIV roughly $2V_P \approx 17.8V$ for capacitor-input half-wave. (d)-(e) surge/repetitive currents require source resistance; with ideal source, surge is unbounded.

Problem 3.96

SPICE simulation problem: build Fig. P3.94, use sinusoidal source, $R = 0.025\Omega$, $C = 0.5F$, diode IS and RS as specified. Measure dc output, ripple, and peak diode current; repeat with $R_S = 0.02\Omega$.

Problem 3.97

Repeat 3.95 but frequency changes. Since $V_r \propto 1/fC$, required capacitance scales inversely with frequency. At $400Hz$, C is $60/400$ times the $60Hz$ value. At $70kHz$, C is $60/70000$ times the $60Hz$ value.

Problem 3.98

Center-tapped full-wave plus Zener regulator. (a) Full-wave rectifier. (b) $V_1 \approx \sqrt{2}(15) - 0.75 = 20.46V$ before ripple average. (c) If regulation holds, $V_O = 15V$. (d) Ripple at V_1 approximate $I/(2fC)$ because full-wave ripple frequency is $2f$. (e) PIV per rectifier diode is about $2V_P$. (f) Negative output version reverses diode/Zener polarities.

Problem 3.99–3.100

Design with ripple limit. For half-wave capacitor filter,

$$C \approx \frac{I_L}{fV_r}, \quad V_r = (\text{ripple fraction})V_{dc}.$$

Transformer rms approximately $(V_{dc} + V_{on} + V_r/2)/\sqrt{2}$. PIV about $2V_P$. Peak repetitive current and surge current require source/diode resistance assumptions; ideal model gives unbounded surge at $t = 0^+$.

Problem 3.101–3.103

Voltage doubler: in steady state ideal no-load output approaches approximately $2V_P$ minus diode drops. Capacitors charge on alternate half cycles. With load, ripple follows $\Delta V \approx I/(fC)$ for each storage capacitor in the doubler path.

3.14-3.16 Full-Wave Rectifiers and Design Tradeoffs

Problem 3.104

Center-tapped full-wave rectifier: output peak approximately $V_P - V_{on}$; ripple frequency is $2f$. For $v_{I,rms} = 18V$, $V_P = 25.46V$, so $V_O \approx 24.46V$ before ripple averaging. Required capacitor for ripple $< 0.25V$:

$$C > \frac{I_L}{2fV_r}, \quad I_L \approx \frac{V_O}{R}.$$

With $R = 0.5\Omega$, $I \approx 48.9A$, $C > \frac{48.9}{120(0.25)} = 1.63F$. PIV for center-tapped full-wave is about $2V_P \approx 50.9V$. Surge current needs source resistance; ideal model gives infinite initial surge.

Problem 3.105

Same as 3.104 with $v_{rms} = 10V$: $V_P = 14.14V$, $V_O \approx 13.14V$; $I \approx 26.3A$ for $R = 0.5\Omega$, so $C > 0.877F$ for $0.25V$ ripple. PIV $\approx 28.3V$.

Problem 3.106

SPICE problem: simulate Fig. P3.104 using $R = 3\Omega$, $C = 22000\mu F$, $v_{rms} = 10V$, $f = 400Hz$, diode model as specified. Compare output dc, ripple, and peak current to the full-wave formulas above.

Problem 3.107–3.108

Repeat 3.99/3.100 for full-wave. Replace f by $2f$ in the ripple formula, so the required C is roughly half the half-wave value for the same load and ripple.

Problem 3.109

From the shown waveforms, only one half-cycle is effectively charging the output, indicating one diode path is open/reversed or one half of the center-tapped secondary is not connected correctly. The circuit is not performing full-wave rectification.

Problem 3.110–3.115

Bridge rectifier: two diodes conduct in series each half-cycle, so

$$V_O \approx V_P - 2V_{on}, \quad V_r \approx \frac{I}{2fC}.$$

PIV per diode is approximately V_P , lower than center-tapped full-wave. Repeat 3.104/3.99/3.100 by using these bridge formulas.

Problem 3.116–3.118

Design tradeoffs: Half-wave: simplest, worst ripple, largest capacitor, poor transformer utilization. Center-tapped full-wave: one diode drop, twice-line ripple, but needs center-tapped transformer and higher PIV. Bridge: no center tap, lower diode PIV, good transformer use, but two diode drops. For low-voltage high-current supplies bridge diode loss is serious; for high-voltage supplies bridge PIV advantage is attractive.

3.17 Dynamic Switching

Problem 3.119

In Fig. P3.119, current immediately after switching is set by the new source voltage and the diode's stored charge/state. At $t = 0^+$, diode current cannot instantly follow the steady reverse current if charge is stored. Estimate forward current before switching and reverse current after switching using resistor law, then storage time approximately

$$t_s \approx \tau_T \ln \left(1 + \frac{I_F}{I_R} \right)$$

(or the specific storage-time formula given in the text).

Problem 3.120–3.122

Simulation exercises. Changing τ_T increases stored charge $Q = \tau_T I_F$ and therefore changes reverse-recovery/storage time. If the diode was forward biased for a long time before switching, more stored charge is present than if the pulse starts at $t = 0$; this explains different observed storage times.

3.18 Photodiodes, Solar Cells, LEDs

Problem 3.123

Solar cell output:

$$I_C = 1 - 10^{-15}(e^{40V_C} - 1).$$

Power delivered is $P = V_C I_C$. Maximum power satisfies

$$\frac{d}{dV}(VI) = I + V \frac{dI}{dV} = 0.$$

$$1 - 10^{-15}(e^{40V} - 1) - 40V(10^{-15})e^{40V} = 0.$$

Solve numerically for V ; then I and $P_{max} = VI$. Short-circuit current $I_{SC} = I(0) = 1A$. Open-circuit voltage solves $I = 0$:

$$V_{OC} = \frac{1}{40} \ln(1 + 10^{15}) \approx 0.864V.$$

Problem 3.124

Series solar cells carry the same current. For a trial current I , each cell voltage is

$$V_{Ci} = \frac{1}{40} \ln \left(1 + \frac{I_{Li} - I}{10^{-15}} \right),$$

where light currents are 1.05, 1.00, and 0.95A. The series current cannot exceed the weakest cell short-circuit current, so $I_{SC} = 0.95A$. Open-circuit voltage is the sum of the three individual V_{OC} values. Maximize $P = I(V_1 + V_2 + V_3)$ numerically.

Problem 3.125

Direct recombination photon energy $E_g = hc/\lambda$. Conveniently,

$$\lambda(\mu m) = \frac{1.24}{E_g(eV)}.$$

Silicon $E_g = 1.12eV$: $\lambda = 1.11\mu m$ infrared. GaAs $E_g = 1.42eV$: $\lambda = 0.873\mu m$ near infrared.

Problem 3.126

Use the same relation $\lambda(\mu m) = 1.24/E_g(eV)$ for each material using its bandgap. Classify color from wavelength: 0.4–0.45 μm violet/blue, 0.45–0.50 blue, 0.50–0.57 green, 0.57–0.59 yellow, 0.59–0.62 orange, 0.62–0.75 red, longer than 0.75 μm infrared. Use the bandgap table specified by the course/text.

Final review checklist

- Abrupt-junction depletion: know ϕ_j , W_d , x_p , x_n , E_{max} .
- Diode equation: $i_D = I_S(e^{v_D/nV_T} - 1)$ and inverse $v_D = nV_T \ln(1 + i_D/I_S)$.
- Drift/diffusion: $J = q(n\mu_n + p\mu_p)E + qD_n dn/dx - qD_p dp/dx$.
- Junction capacitance: $C_j = C_{j0} \sqrt{\phi_j/(\phi_j + V_R)}$.
- Rectifiers: capacitor ripple is set by load current, capacitance, and recharge frequency.
- Diode circuit models: ideal, CVD, mathematical model; always verify assumed diode states.